10/19/2026

Last time!

Thm. The units in $\mathbb{Z}[\sqrt{2}]$ are

$$\pm \varepsilon_0^n = \pm (1+\sqrt{2})^n, \quad n \in \mathbb{Z}.$$

If $\varepsilon_n = \pm \varepsilon_0^n = x_n + y_n\sqrt{2}$, where $x_n, y_n \in \mathbb{Z}$, then

$$x_n^2 - 2y_n^2 = N(\pm\varepsilon_n) = (N(\varepsilon_0))^n = (-1)^n.$$

So even values of n gives all $x, y \in \mathbb{Z}$ satisfying

$$x^2 - 2y^2 = 1,$$

and odd values of n gives all $x, y \in \mathbb{Z}$ satisfying

$$x^2 - 2y^2 = -1.$$

Units in $\mathbb{Z}[\sqrt{d}]$ and Pell's equations $x^2 - dy^2 = \pm 1$.

Unit $\varepsilon = x + y\sqrt{d}$ in $\mathbb{Z}[\sqrt{d}]$ $\longrightarrow$ Solution $(x, y) \in \mathbb{Z}^2$ to $x^2 - dy^2 = \pm 1$.

$\longleftarrow$

Suppose $d > 1$ is not a square.

1. $x^2 - dy^2 = 1$ always has a fundamental solution

$\varepsilon_0 = x_0 + y_0\sqrt{d}$, the least unit > 1, such that every solution

$\varepsilon = x + y\sqrt{d}$ is one of $\pm \varepsilon_0^n$ $(n \in \mathbb{Z})$.

2. $x^2 - dy^2 = -1$ may not have a solution, but when it does, it has a fundamental solution $\varepsilon_0 = x_0 + y_0\sqrt{d}$, (unit)

the least unit > 1, such that every solution $\varepsilon = x + y\sqrt{d}$ to $x^2 - dy^2 = (-1)^n$ is one of $\pm \varepsilon_0^n$ $(n \in \mathbb{Z})$.

Q: How to find the fundamental solution $\varepsilon_0 = x_0 + y_0\sqrt{d}$?

Idea 1. Take positive integers y s.t.

$$\pm 1 + dy^2 = \square$$

E.g. Find the fundamental solution to $x^2 - 7y^2 = 1$.

Look at $1 + 7y^2 = x^2$: $y = 1, 2$ ✗

$y = 3$, $x = 8$.

So the fundamental solution is $(8, 3)$.

Hence all the solutions are

$$\pm (8+3\sqrt{7})^n \quad , \quad n \in \mathbb{Z}.$$

Idea 2. Use "continued fractions".

Rational numbers: $\frac{7}{4} = 1 + \frac{3}{4} = 1 + \frac{1}{\frac{4}{3}}$

$= 1 + \frac{1}{1 + \frac{1}{3}}$

$= [1; 1, 3]$

Quadratic irrationals:

$\sqrt{2} = 1 + (\sqrt{2} - 1) = 1 + \frac{1}{\frac{1}{\sqrt{2}-1}} = 1 + \frac{1}{\frac{\sqrt{2}+1}{(\sqrt{2}-1)(\sqrt{2}+1)}}$

$= 1 + \frac{1}{\sqrt{2}+1} = 1 + \frac{1}{2 + (\sqrt{2} - 1)}$

$= 1 + \frac{1}{2 + \frac{1}{\sqrt{2}+1}} = 1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \cdots}}}} = [1; \overline{2}]$.

$$\frac{1+\sqrt{5}}{2} = 1 + \frac{\sqrt{5}-1}{2} = 1 + \frac{1}{\frac{2}{\sqrt{5}-1}} = 1 + \frac{1}{\frac{2(\sqrt{5}+1)}{(\sqrt{5}-1)(\sqrt{5}+1)}}$$

Golden ratio

$$= 1 + \frac{1}{\frac{\sqrt{5}+1}{2}} = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1+\cdots}}} = [1; \overline{1}].$$

A (simple) continued fraction has the form

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cdots}}} = [a_0; a_1, a_2, a_3, \ldots].$$

$a_0, a_1, a_2, \ldots, \geq 0$

Transcendental numbers !

$$e = 2.71828\ldots = 2 + \cfrac{1}{1 + \cfrac{1}{2 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{4 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{6 + \cfrac{1}{1 + \cdots}}}}}}}}}$$

$$= [2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, 1, 1, 10, 1, \ldots]$$

non-periodic

If $d > 1$ is not a square and $x, y > 0$ is a solution to $x^2 - dy^2 = 1$ (or $x^2 - dy^2 = -1$), then

$$\frac{x}{y} = a_0 + \cfrac{1}{a_1 + \cdots + \cfrac{1}{a_k}} = [a_0; a_1, \ldots, a_k], \text{ which is}$$

a truncation of $\sqrt{d} = [a_0; a_1, \cdots, a_k, \cdots]$

Periodic

$= [a_0; \overline{a_1, \cdots, a_\ell}]$.

Ex. $x^2 - 13y^2 = 1$.

$\sqrt{13} = 3 + (\sqrt{13} - 3)$

$\frac{1}{\sqrt{13}-3} = \frac{\sqrt{13}+3}{(\sqrt{13}-3)(\sqrt{13}+3)} = \frac{\sqrt{13}+3}{4} = 1 + \frac{\sqrt{13}-1}{4}$

$\frac{4}{\sqrt{13}-1} = \frac{\sqrt{13}+1}{3} = 1 + \frac{\sqrt{13}-2}{3}$

$\frac{3}{\sqrt{13}-2} = \frac{\sqrt{13}+2}{3} = 1 + \frac{\sqrt{13}-1}{3}$

$\frac{3}{\sqrt{13}-1} = \frac{\sqrt{13}+1}{4} = 1 + \frac{\sqrt{13}-3}{4}$

$\frac{4}{\sqrt{13}-3} = \sqrt{13}+3 = 6 + (\sqrt{13}-3)$

So $\sqrt{13} = [3; \overline{1, 1, 1, 1, 6}]$

Rmk. The continued fraction expansion of $x \in \mathbb{R}$ is finite iff $x \in \mathbb{Q}$. It is periodic iff x is a quadratic rational.

Will use this to solve $x^2 - 13y^2 = 1$.